

OOS — Master Report

Period: Aug 11, 12, 15, 19, 20 (2026) — 5 clean analysis days **Excluded:** Aug 13-14 (Spark/PUBLICADOR outage) **Date analyzed:** 2026-08-25 **Script:** `scripts/classify_oos.py`

Executive Summary

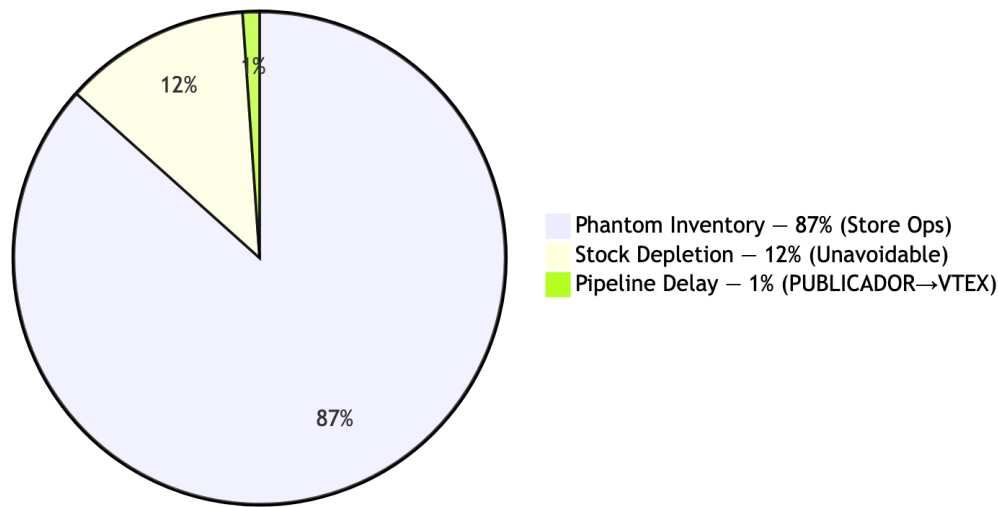
Across 5 analysis days in August 2026, we classified **49,779 out-of-stock order lines** from **1.92M total order lines** to determine **why customers see items as available on jumbo.cl but the picker finds an empty shelf**.

Key Numbers

Metric	Value
OOS Rate	2.59% average (improving: 2.97% → 2.23% over the period)
Perfect orders	46.8% — every item picked exactly as ordered, no substitutions
Orders affected by OOS	28.8% of orders have ≥1 missing item (not delivered)
Orders with substitution only	~17.8% — item replaced, no missing items
Found Rate (strict)	94.1% — item-level, picked exactly as ordered
Found Rate (all positive)	97.8% — including substitutions, partials, and picker-added items
Typical OOS impact per order	1 item missing out of ~22 (93% of the order delivered)
Total OOS exposure	267.5M CLP across 5 days (~53.5M CLP/day)

Root Cause Attribution

Why Are Items Out of Stock? (49,779 lines)



Root Cause	Share	Description	Owner
Phantom inventory	86.6%	HANA/PUBLICADOR reports stock available, shelf is empty. Identical rate in dark stores (89.4%) and regular stores (89.0%) — not caused by shared inventory with walk-in customers. 78.7% of phantom items had zero prior ecommerce picks.	Store Operations
Stock depletion	12.3%	Stock legitimately ran out between PUBLICADOR batches. Customer ordered while item was still published, but stock depleted before picker arrived.	Unavoidable
Pipeline delay	0.7%	PUBLICADOR sent NO PUBLICAR but VTEX hadn't processed the update yet, or item blocked all day but still orderable.	PUBLICADOR→VTEX Pipeline
Catalog gap	0.4%	Item+store pair not found in any PUBLICADOR batch. New items or mapping gaps.	Catalog Ops

Phantom Diagnosis — Why Is the Shelf Empty?

We tracked 25,999 unique (SKU, store) pairs that went OOS across 5 days and tested every hypothesis:

Hypothesis	Test	Result	Verdict
Prior ecommerce orders depleted stock	Checked if earlier picked orders consumed the PUBLICADOR-reported qty before the OOS order	Only 1.1% fully depleted. 78.7% of phantom items had zero prior picks — shelf was already empty when the first order came in.	Ruled out
Walk-in customers took the stock	Compared dark stores (no walk-in traffic) vs regular stores (shared inventory with physical customers)	Phantom rate nearly identical: dark 89.4% vs regular 89.0%. If walk-ins were the cause, regular stores should be much higher.	Ruled out
High-demand SKUs get depleted faster	Correlated order volume per SKU+store with phantom rate	Phantom rate flat at 85-89% across all volume buckets — even single-order SKUs are phantom 88.7% of the time.	Ruled out
Persistent shelf gap (HANA overestimates stock)	Tracked same (SKU, store) pairs across 5 days	77.6% transient (1 day only), but 360 chronic pairs phantom every day. 17.1% of phantom items stay phantom the next day.	Primary cause

Conclusion: The root cause is **shelf-level inventory inaccuracy**. SAP/HANA reports stock on hand, PUBLICADOR correctly publishes it, VTEX correctly shows it to the customer — but the physical item is not on the shelf. This is not a pipeline problem, not a demand problem, and not a shared-inventory problem. It is a store operations problem: cycle counting gaps, shrinkage, misplacement, or receiving errors.

What This Means

1. **The PUBLICADOR→VTEX pipeline is not the problem.** Less than 1% of OOS is caused by propagation delays. Pipeline optimization has diminishing returns.
2. **Dark stores are the highest-value target.** Dark stores (J4xx/J5xx) have 3.3× the CLP exposure (~41M CLP/day) despite the same phantom rate as regular stores. Fixing inventory accuracy in dark stores yields the highest ROI.

3. **Most phantom cases are transient (1-day).** 77.6% of phantom (SKU, store) pairs appear on only 1 day. But 360 chronic pairs (phantom 4-5 consecutive days) are persistent shelf gaps that should trigger automatic cycle counts.
4. **Top offending categories: meat, produce, fruit.** Items with high physical shrinkage and inventory count inaccuracy dominate the phantom list.

Recommendations

#	Action	Target	Expected Impact
1	Cycle count program for chronic phantom pairs	360 pairs with 4-5 day persistence	Eliminate ~20M CLP/5-day chronic exposure
2	Dark store inventory audit	J4xx/J5xx stores (18,104 OOS pairs)	Reduce 204.6M CLP/5-day dark store OOS
3	PUBLICAR threshold adjustment	Publish only when HANA qty $\geq$ N (not qty $\geq$ 1)	Prevent ~4,244 recurring phantom pairs
4	Fix negative/zero stock publishing	A1+A2: 2,098 lines published with qty $\leq$ 0	Quick win — PUBLICADOR rule change

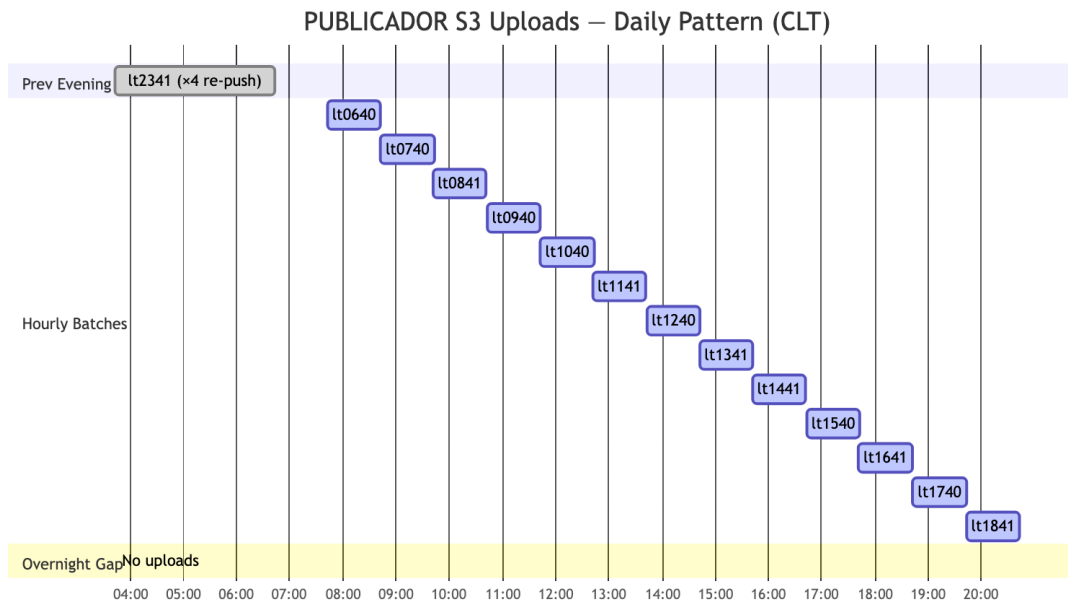
Bottom line: OOS rate is improving (2.97% $\rightarrow$ 2.23%), and the pipeline is healthy. The next lever is store-level inventory accuracy — specifically, closing the gap between what HANA reports and what is physically on the shelf. Dark stores and chronic phantom pairs are the highest-ROI targets.

1. Data Sources

Source	Description	Per Day
Orders	Redshift <code>jumbo_bo.io_items</code> + <code>vw_master_pickers_data</code>	340-495K lines
PUBLICADOR	S3 CSV files, semicolon-delimited, ~1.19M rows each	18 files
VTEX Events	SQS drain JSONL (for future pipeline delay measurement)	3-14M events
SKU Mapping	<code>vtex_sap_mapping.csv</code> (dim_surtidos) — VTEX skuld to SAP refid	74,804 rows

PUBLICADOR S3 Upload Schedule

PUBLICADOR generates CSV snapshots from SAP/HANA data. Each file contains ~1.19M (item, store) pairs with stock quantities and publish/block decisions.

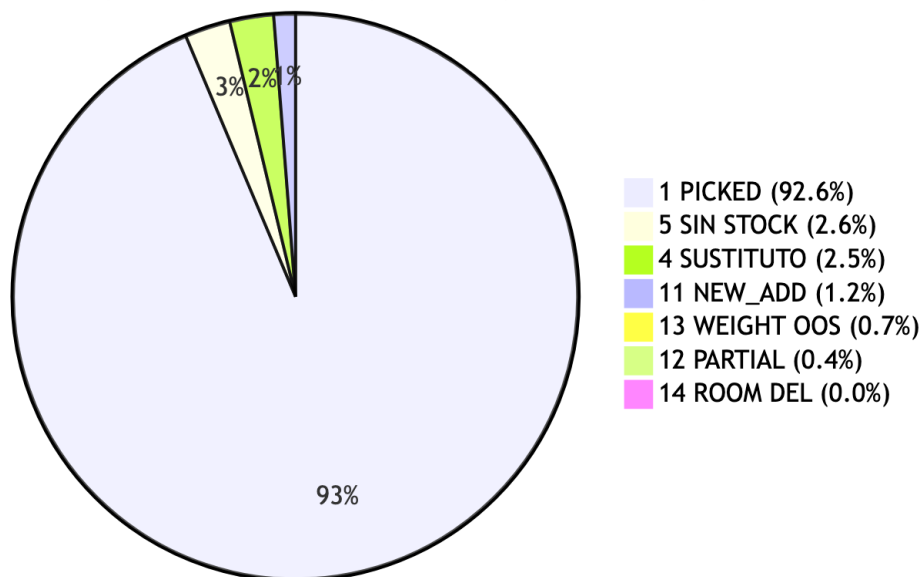


Files/day: 18 (1 previous evening × 4 re-pushes + 13 hourly + 1 evening)
 Window: 03:43 CLT (overnight re-push) → 19:43 CLT (last batch)
 Gap: 19:43 → 03:43 (8-hour overnight gap, no new data)
 Timing: PUBLICADOR Load_Time + ~62 min = S3 upload time

2. Order Volume & Status Breakdown

5-Day Aggregate (1,921,770 lines)

Order Lines by Status — 5-Day Total (1.92M lines)



Status	Name	Lines	%	CLP (M)	Included in OOS?
1	PICKED	1,779,624	92.6%	8,105.2	No
5	VALIDATE_NO_STOCK	49,598	2.6%	257.3	Yes
4	VALIDATE_SUSTITUTE	47,640	2.5%	241.6	No
11	NEW_ADD	23,590	1.2%	115.9	No (excluded)
13	FREE_PICKING_VALIDATE	12,802	0.7%	198.4	Flagged
12	PARTIAL_ADD_VALIDATE	8,335	0.4%	76.2	No
14	ROOM_DELETION	181	0.0%	0.2	Yes

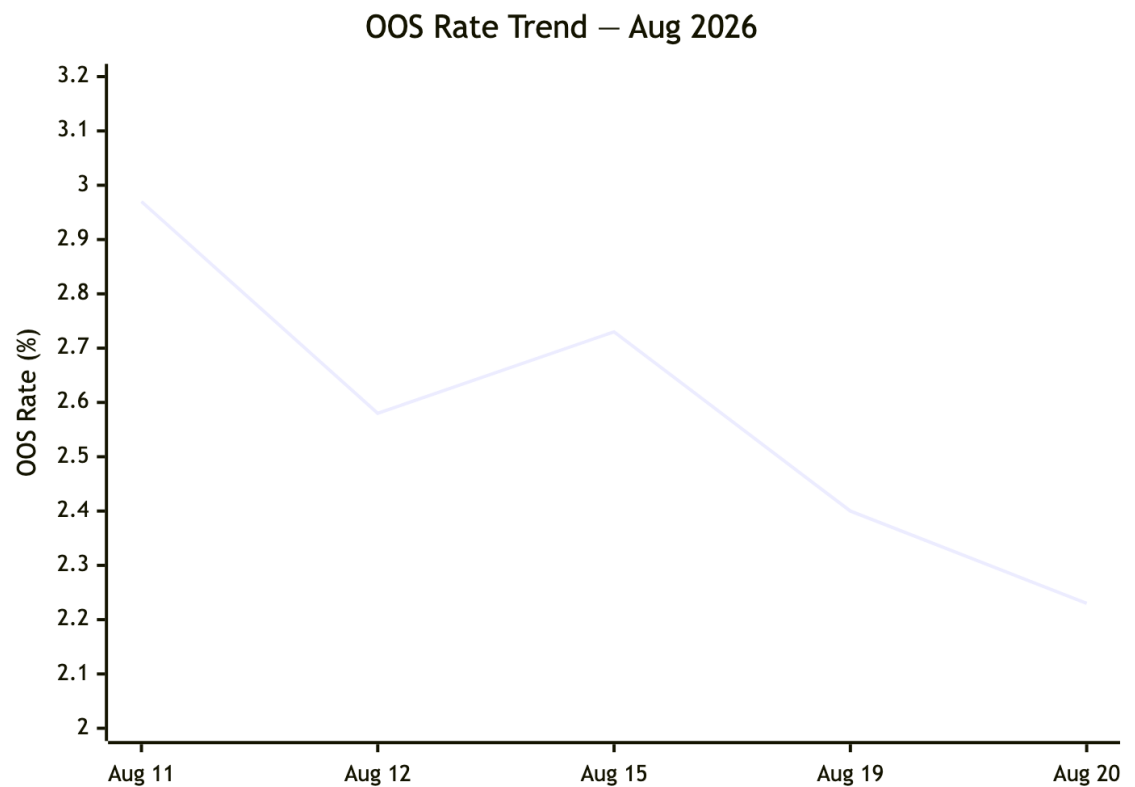
Formulas: - **OOS Rate** = (status 5 + 14) / total lines - **OOS Broad** = (status 3 + 5 + 13 + 14) / total lines — includes pre-validation and weight-item OOS - **Found Rate (strict)** = status 1 / (total - status 11) — picked exactly as ordered - **Found + Substitute** = (status 1 + 4) / (total - status 11) — includes substitutions - **Found + Sub + Partial** = (status 1 + 4 + 12) / (total - status 11) — includes partial fills - **Found + Sub + Partial + Added** = (status 1 + 4 + 11 + 12) / (total - status 11) — all positive outcomes - Status 11 excluded from denominator: picker-added items aren't customer orders - **Status 13** is effectively OOS (pickingqty=0) but uses a separate code for weight items. 12,802 lines (198.4M CLP) flagged but not classified.

3. Multi-Day Summary

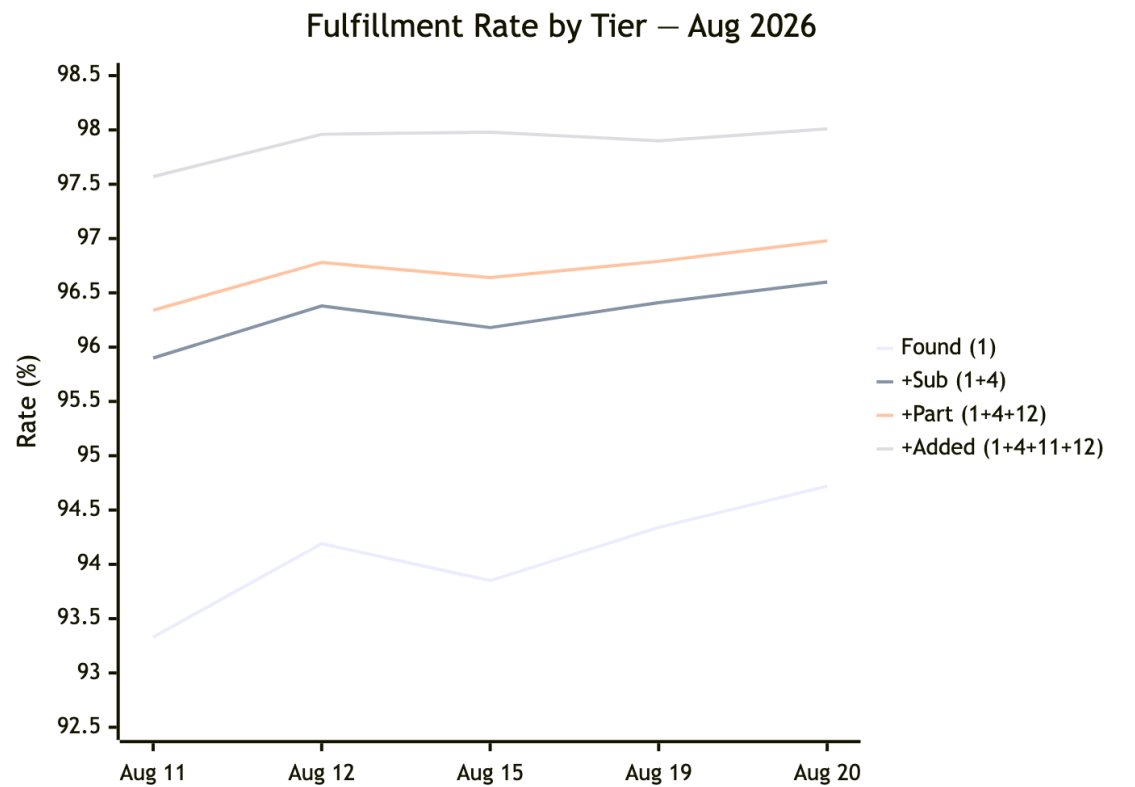
Day	Lines	Orders	OOS	OOS Rate	Found (1)	+Sub (1+4)	+Partial (1+4+12)	+Added (1+4+11+12)
Aug 11	361,215	22,145	10,737	2.97%	93.33%	95.90%	96.34%	97.57%
Aug 12	343,503	21,267	8,875	2.58%	94.19%	96.38%	96.78%	97.96%
Aug 15	495,287	27,628	13,518	2.73%	93.85%	96.18%	96.64%	97.98%
Aug 19	341,809	21,422	8,187	2.40%	94.34%	96.41%	96.79%	97.90%
Aug 20	379,956	22,992	8,462	2.23%	94.72%	96.60%	96.98%	98.01%
Total	1,921,770	115,454	49,779	2.59%	94.11%	96.11%	96.55%	97.81%

Aug 15 is the largest day (Friday effect — 37% more orders than average). Substitutions add ~2.2-2.6pp to the found rate. Partials add ~0.4pp. Picker-added items add ~1.0pp.

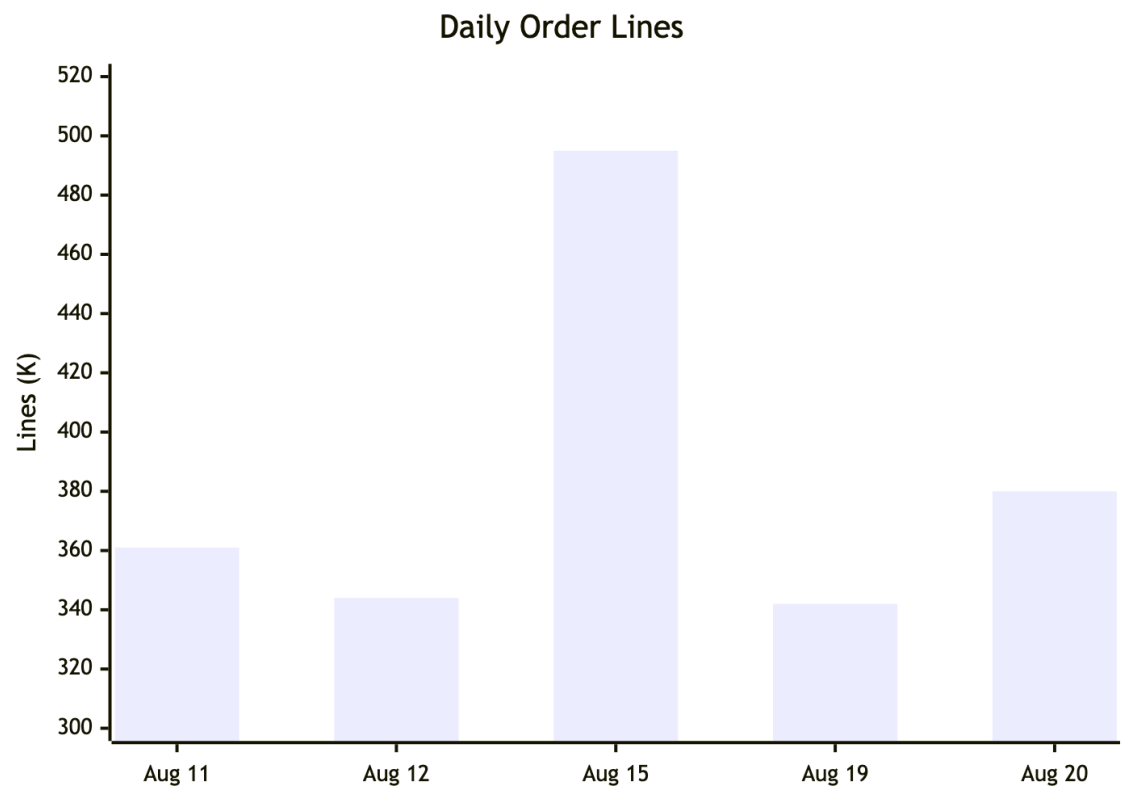
OOS Rate Trend



Fulfillment Tier Trend



Order Volume by Day



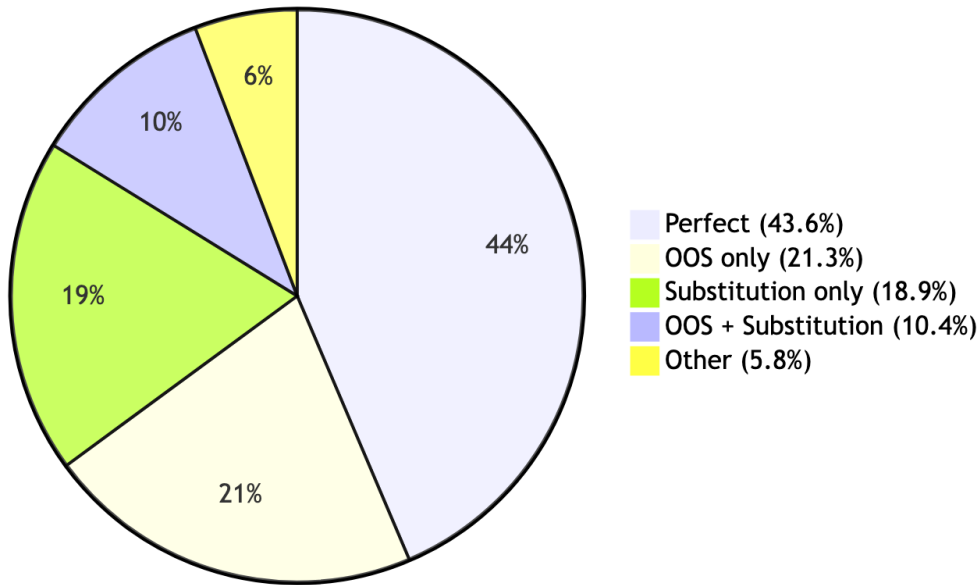
4. Customer Impact — Order Level

Order Outcome Profile

Not all orders are affected equally. Most orders deliver successfully; OOS impacts a subset.

Outcome	Aug 11	Aug 12	Aug 15	Aug 19	Aug 20
Perfect (100% picked)	43.6%	48.1%	42.6%	49.2%	50.5%
Substitution only	18.9%	17.7%	18.9%	17.3%	16.3%
OOS only (no sub)	21.3%	20.1%	21.6%	18.9%	18.2%
OOS + substitution	10.4%	8.4%	10.3%	7.5%	7.2%
Other (partial/weight)	5.8%	5.8%	6.6%	7.2%	7.7%

Order Outcomes – Aug 11 (22,145 orders)



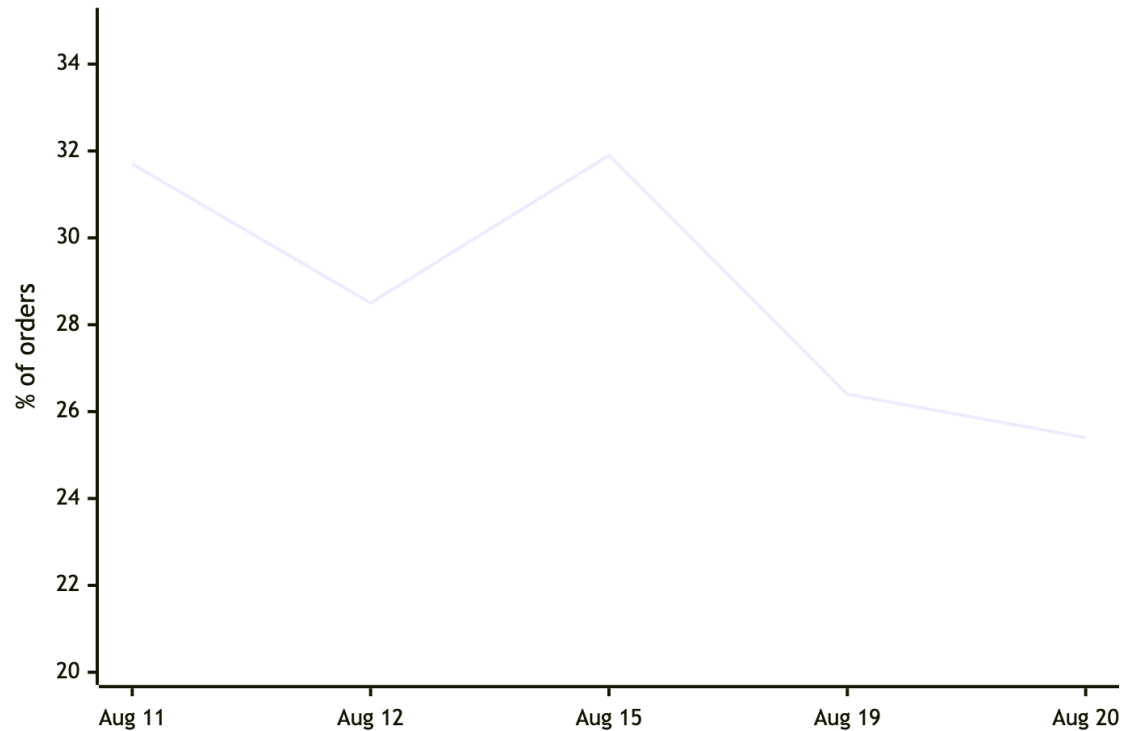
OOS-Affected Orders

Metric	Aug 11	Aug 12	Aug 15	Aug 19	Aug 20
Total orders	22,145	21,267	27,628	21,422	22,992
OOS-affected orders	7,025	6,053	8,803	5,658	5,847
% affected	31.7%	28.5%	31.9%	26.4%	25.4%
Avg items/order	21.4	21.6	22.8	21.2	22.0
Avg OOS items/order	1.5	1.5	1.5	1.4	1.4
Avg % OOS per order	7.1%	6.8%	6.7%	6.8%	6.6%

OOS Severity per Order

Severity	Aug 11	Aug 12	Aug 15	Aug 19	Aug 20
1 OOS item	66.8%	69.8%	66.5%	71.1%	70.9%
2+ OOS items (< 50%)	32.9%	29.9%	33.3%	28.6%	28.8%
50-99% OOS	0.6%	0.5%	0.4%	0.4%	0.4%
100% wipeout	0.2%	0.2%	0.1%	0.2%	0.1%

% Orders Affected by OOS — Trending Down



Typical impact: 1 out of ~22 items missing. 67-71% of affected orders have only 1 missing item. Total wipeouts (100% OOS) are 7-17 per day (0.1-0.2%). Items confirmed not delivered: `pickingqty = 0` for all 49,779 OOS lines.

5. Classification Algorithm (PUBLICADOR-Based)

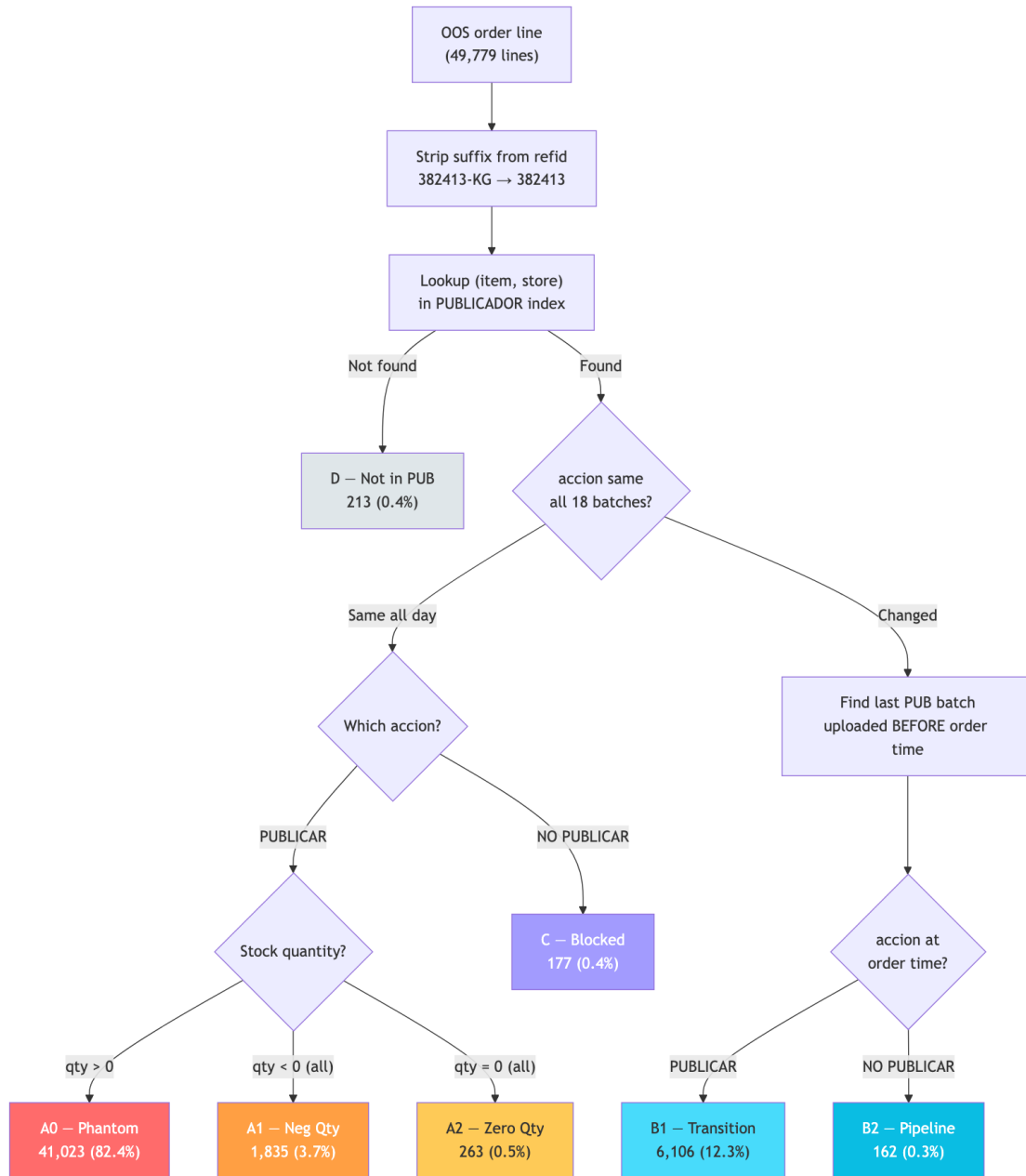
Question

For each OOS order line (status 5 or 14): **why did the customer see this item as in-stock on jumbo.cl?**

Join Key

```
Orders refid: "382413-KG" → strip -KG/-PAK/-CS → "382413"
PUB Item_Id:  "382413"   → direct match
Store:        id_tienda "J411" = PUB Location_Id "J411"
```

Flowchart



6. Classification Categories

A0 — Phantom Inventory (82.4% of OOS)

PUBLICADOR says PUBLICAR with positive stock **all day**. VTEX correctly shows item in-stock. Customer orders. But the physical shelf is empty.

Example	refid	Store	Order Time	PUB qty	caso
Agua de Colonia Barzelatto 500 cc	273608	J955	00:25	12	JERARQUIA CON STOCK REGULAR
Summer Mix Dole 150 g	920715	J955	00:25	4	REGLA DDS

A1 — Phantom Negative Qty (3.7%)

PUBLICADOR says PUBLICAR despite **negative** stock in every batch.

Example	refid	Store	Order Time	PUB qty	caso
Pan Marraqueta Wholegrain Granel	1962049-KG	J592	00:53	-0.5	LISTADO FIJO SIN STOCK
Ramo de Mini Rosas 6 un.	989736	J403	00:53	-86	PRODUCTO TIENDA MADRE

A2 — Phantom Zero Qty (0.5%)

PUBLICADOR says PUBLICAR despite **zero** stock in every batch.

Example	refid	Store	Order Time	PUB qty	caso
Palta Reina 1 un.	1359541	J760	06:31	0	LISTADO FIJO SIN STOCK
Croissant Cuisine & Co 3 un.	2018277	J410	08:59	0	PRODUCTO TIENDA MADRE

B1 — Transition, PUBLICAR at Order (12.3%)

Item's accion **changed** during the day. At order time, the latest PUBLICADOR batch still said PUBLICAR.

Example	refid	Store	Order Time	PUB qty	caso
Chuleta Centro Cong Super Cerdo 750 G	1712073	J521	00:13	4	JERARQUIA PRODUCCION
Agua Saborizada Benedictino Manzana 1.5 L	1820043	J760	00:01	19	JERARQUIA PRODUCCION

B2 — Transition, NO PUBLICAR at Order (0.3%)

PUBLICADOR had **already** sent NO PUBLICAR before the order, but the customer was still able to order. Possible pipeline propagation delay.

Example	refid	Store	Order Time	PUB qty	caso
Yogurt Soprole Proteina Natural 155 g	1775576005	J775	00:42	-28	JERARQUIA CON STOCK REGULAR
Rucula Vegus Organica 200 g	728565	J512	08:44	0	REGLA DDS

C — Blocked All Day, VTEX Stale (0.4%)

PUBLICADOR said NO PUBLICAR in **every** batch all day. VTEX never processed the block.

Example	refid	Store	Order Time	PUB qty	caso
Chocolates Rolls Crispy 100 g	1926388	J408	10:44	20	PRODUCTO TIENDA MADRE
Saborizante Cola Cao Chocolate 400 g	265912	J411	13:48	0	JERARQUIA CON STOCK REGULAR

D — Not in PUBLICADOR (0.4%)

Item+store pair not found in any of the 18 PUBLICADOR batches. New items or catalog gaps.

Example	refid	Store	Order Time
Palta Hass Pote 2 un.	2032557	J988	16:58
Hummus Perfect Choice Tradicional 220 g	1838040	J411	16:49

7. Results by Day

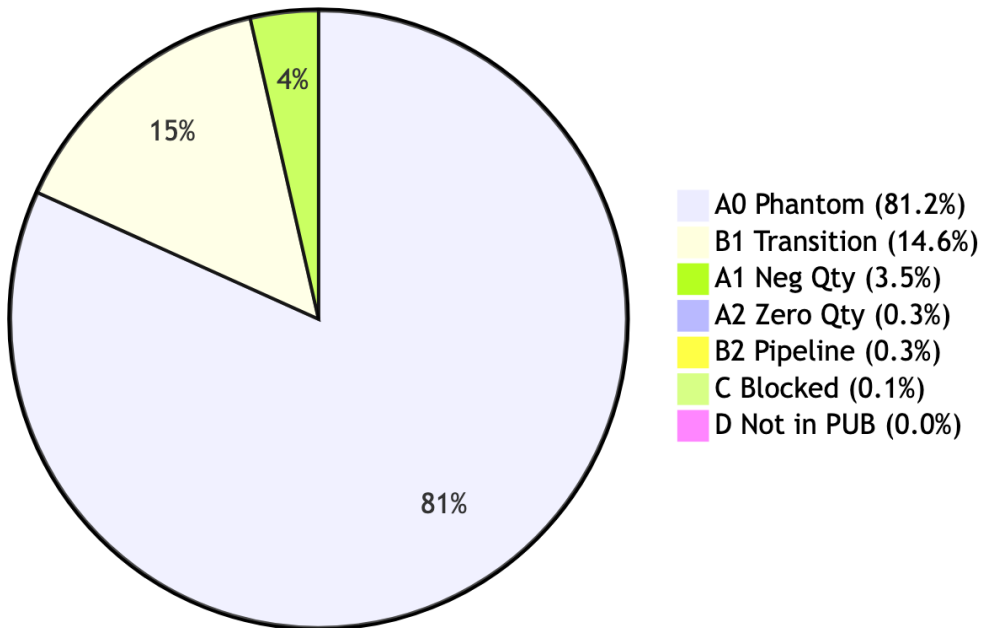
Classification Counts

Category	Aug 11	Aug 12	Aug 15	Aug 19	Aug 20	Total
A0 Phantom	8,716	7,196	11,168	6,831	7,112	41,023
A1 Neg Qty	380	369	485	301	300	1,835
A2 Zero Qty	33	31	46	69	84	263
B1 Transition	1,569	1,218	1,479	912	928	6,106
B2 Pipeline	27	24	56	43	12	162
C Blocked	7	19	117	24	10	177
D Not in PUB	5	18	167	7	16	213
Total OOS	10,737	8,875	13,518	8,187	8,462	49,779

Per-Day Classification Charts

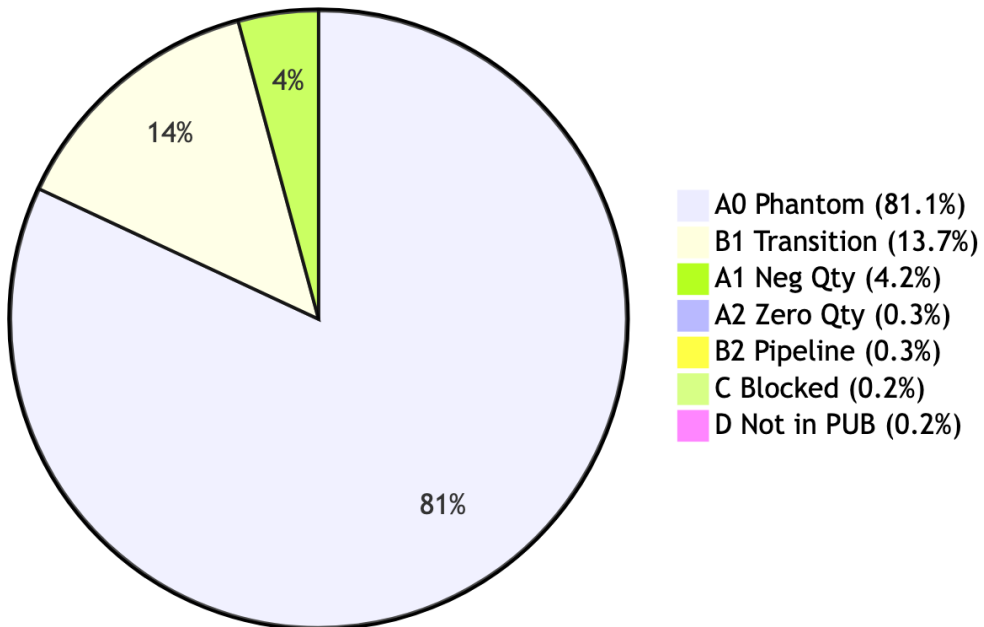
Aug 11 (10,737 OOS lines)

Aug 11 — OOS Classification



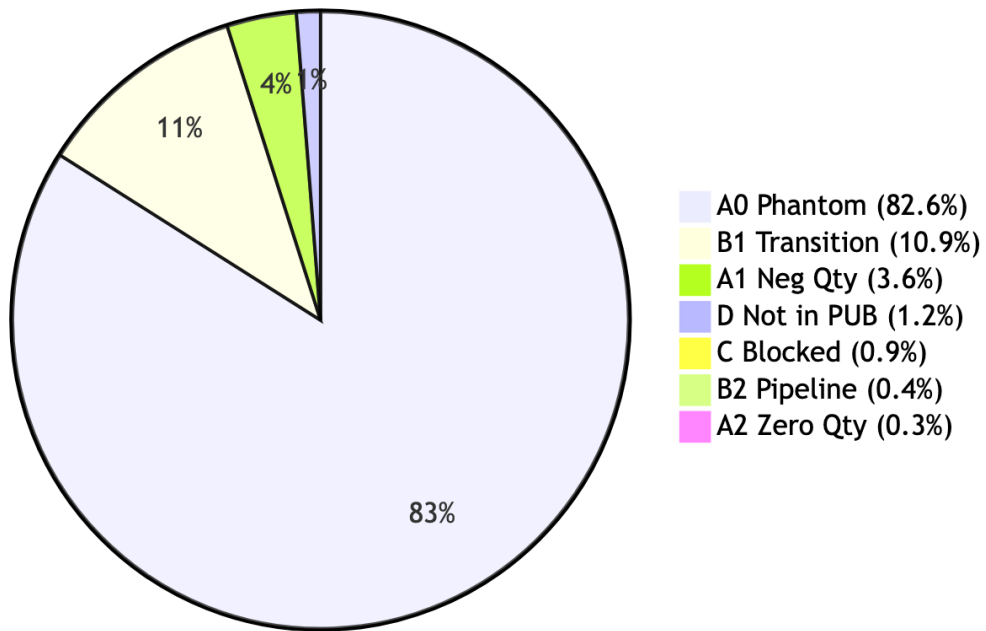
Aug 12 (8,875 OOS lines)

Aug 12 — OOS Classification



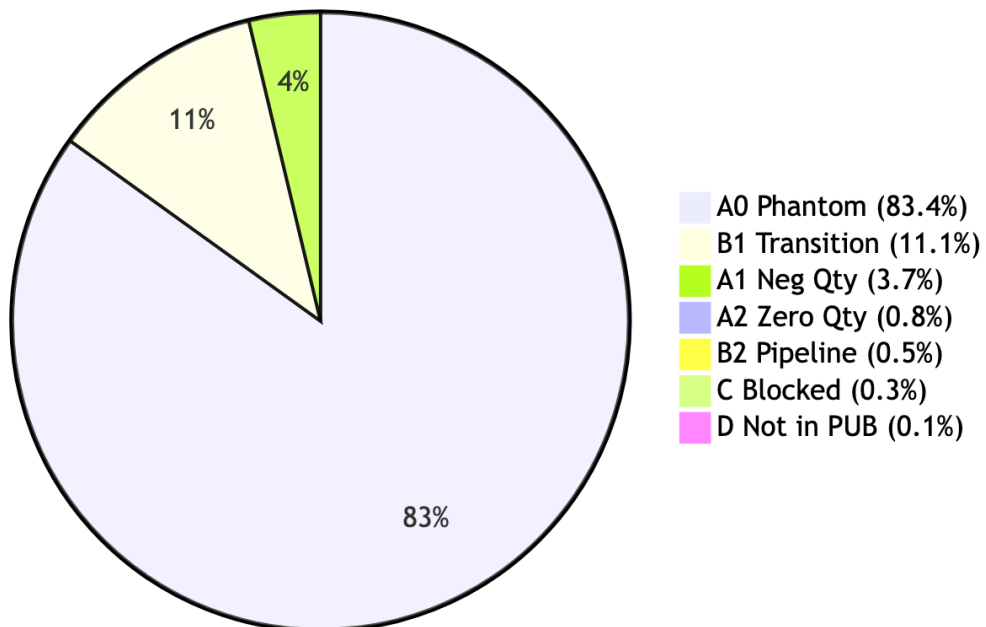
Aug 15 (13,518 OOS lines)

Aug 15 — OOS Classification (Friday)



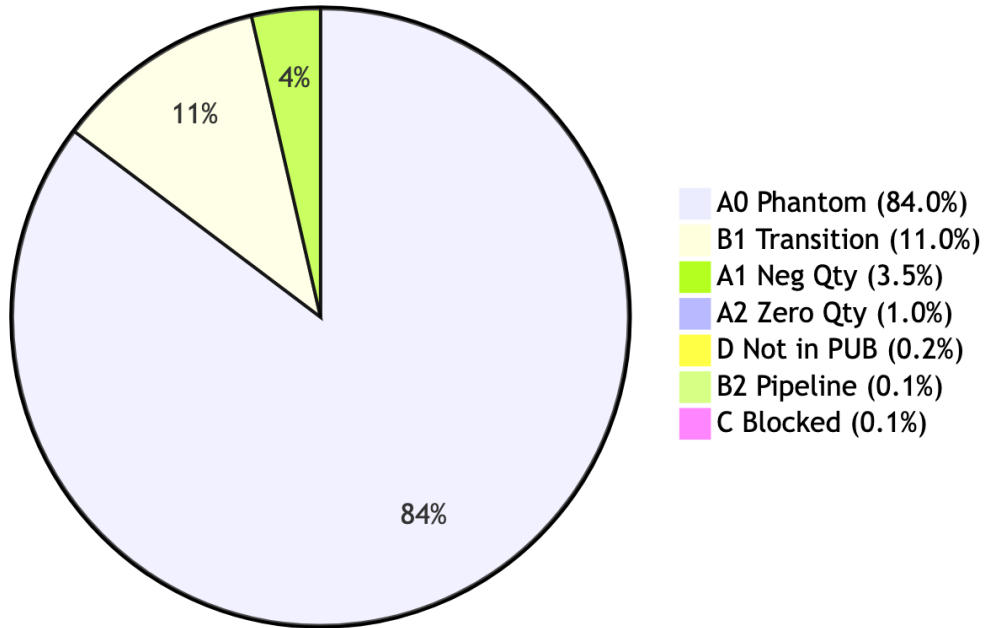
Aug 19 (8,187 OOS lines)

Aug 19 — OOS Classification



Aug 20 (8,462 OOS lines)

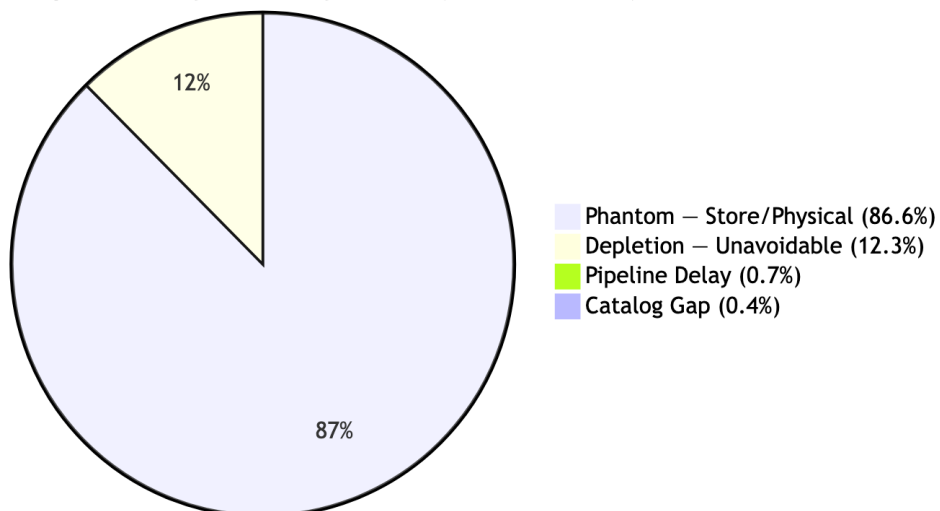
Aug 20 – OOS Classification



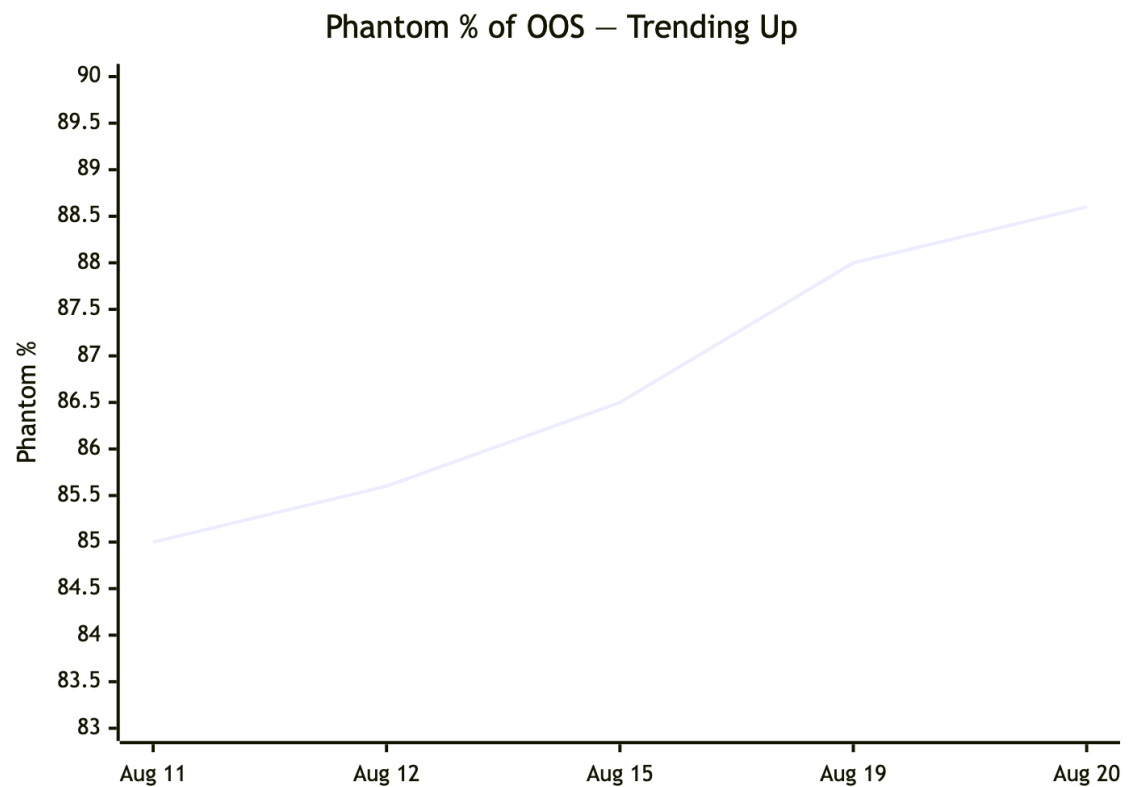
Responsibility Split

Responsibility	Aug 11	Aug 12	Aug 15	Aug 19	Aug 20	Avg
Phantom (store)	85.0%	85.6%	86.5%	88.0%	88.6%	86.6%
Depletion (unavoidable)	14.6%	13.7%	10.9%	11.1%	11.0%	12.3%
Pipeline delay	0.3%	0.5%	1.3%	0.8%	0.3%	0.7%
Catalog gap	0.0%	0.2%	1.2%	0.1%	0.2%	0.4%

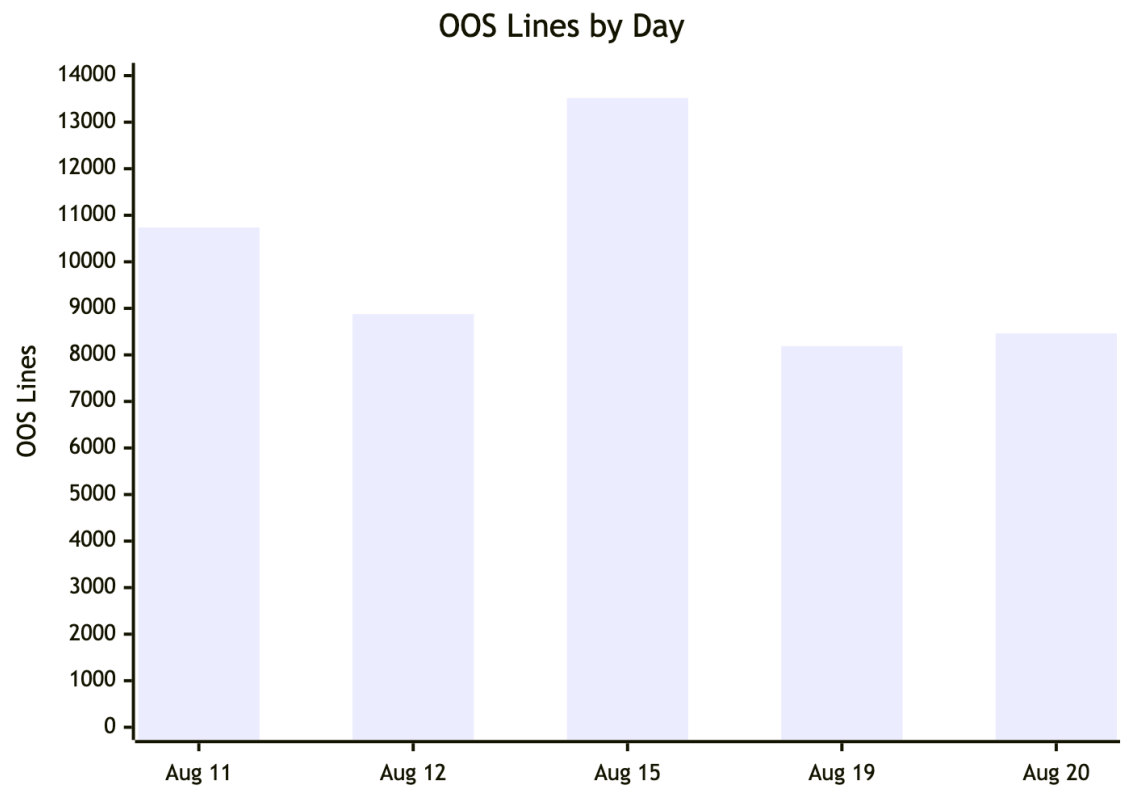
OOS by Responsibility – 5-Day Total (49,779 lines)



Phantom Rate Trend (% of OOS that is Phantom)



OOS Lines by Day (Stacked by Category)



8. Top Phantom Items Across All Days

Item	refid	Total CLP (K)	OOS Lines	Days
Uva Verde Granel	1580859-KG	2,112	240	5/5
Carne Molida Vacuno 4% Grasa 500 g	1998017	2,058	161	5/5
Carne Molida Congelada 4% Cuisine & Co	2073560	1,906	153	4/5
Palta Hass Extra Chilena (2 un.)	506862-KG	1,703	351	5/5
Carne Molida Vacuno 7% Grasa 500 g	1998019	1,372	116	5/5
Reineta Filete Fresca Granel	436869-KG	1,165	69	5/5
Punta de Ganso Al Vacio kg	1462107-KG	921	37	5/5
Posta Negra Desgrasada Bandeja 750 g	1998537-KG	850	44	4/5
Bistec de Asiento Bandeja 300 g	1995694-KG	841	86	4/5
Frutilla Pote 300 g	1497672	835	215	5/5

Top offenders are **meat, produce, and fruit** — categories with high physical shrinkage and inventory count inaccuracy.

9. Key Findings

- 87% of OOS is phantom inventory** — PUBLICADOR correctly instructs VTEX to show the item as in-stock (PUBLICAR + positive qty), but the physical shelf is empty. This is a store operations problem: system stock count does not match reality.
- Pipeline delay is negligible (<1%)** — Only 339 of 49,779 OOS lines (0.7%) are attributable to PUBLICADOR-to-VTEX propagation delay. The pipeline is not the bottleneck.
- ~4% of phantoms have negative or zero stock (A1 + A2)** — PUBLICADOR publishes items despite stock_umb being negative or zero. These are PUBLICADOR rule issues: negative stock should trigger NO PUBLICAR. Fixing this would prevent ~2,098 OOS lines across 5 days.
- Status 13 is uncounted OOS** — 12,802 additional lines across 5 days with pickingqty=0 use status 13 instead of 5/14. These are weight items (meat, produce). Including them raises OOS rate by ~0.5-0.8pp per day.
- OOS rate is improving** — 2.97% (Aug 11) to 2.23% (Aug 20), consistent decline. Found rate improving in parallel (93.3% to 94.7%).
- Phantom rate is increasing** — While absolute OOS drops, the phantom share grows from 85.0% to 88.6%. As pipeline-related OOS gets fixed, store-level inventory accuracy becomes the dominant lever.

10. Output Files

File	Description
scripts/classify_oos.py	Reusable classification script
analysis/aug_analysis/{tag}/oos_classification_{tag}.csv	Per-line classification (5 files)
analysis/aug_analysis/{tag}/orders_{tag}.csv	Full order lines (5 files)
analysis/aug_analysis/{tag}/publicador/	18 PUBLICADOR CSV files per day